

How ERROR CORRECTING CODES Help In Communication And Storage



Atul M. Kulkarni is M.Tech in communication engineering from National Institute of Technology (NIT), Karnataka. Currently, he is lecturer at Dr Babasaheb Ambedkar Technological University, Maharashtra. This article has been written under the guidance of Dr Prashantha Kumar H., assistant professor at NIT, Karnataka

No storage media like CDs, DVDs, flash memories, solid-state devices/memories or telecommunication technologies like mobile communication, satellite communication, WiMAX, digital audio/video broadcasting (DVB), deep Space communication is perfect, and all of these experience some amount of errors. To prevent re-reading or re-transmission of

information every time an error occurs, storage and communication systems are embedded with error correcting code (ECC) technology.

History of coding theory

Pioneers of coding theory, Claude Elwood Shannon (1916-2001) and Richard Hamming (1915-1998), were colleagues at Bell Laboratories. Shannon's channel coding theorem proves that if the information transmission rate is less than the channel capacity, it is possible to design an ECC with almost error-free information transmission. American mathematician Hamming pioneered this field and invented the first ECC in 1950. It is known as (7, 4) Hamming code.

There are various error-detection and correction techniques, which are suitable with specific noisy channels/media, like random error or burst error or multi-path distortion or channel effects. Error control

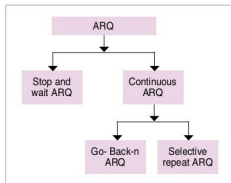


Fig. 2: Classification of ARQ



Fig. 1: Pioneers of coding theory Claude Elwood Shannon (top) and Richard Hamming (bottom)

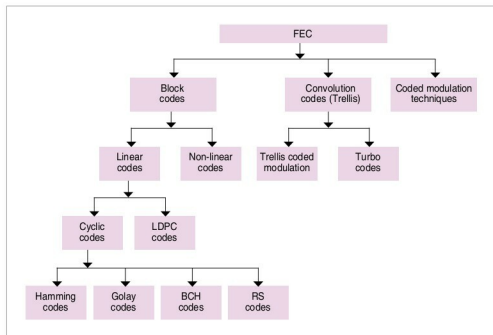


Fig. 3: Classification of FEC